Chance-constrained model predictive control for green and grid-connected hydrogen refuelling stations

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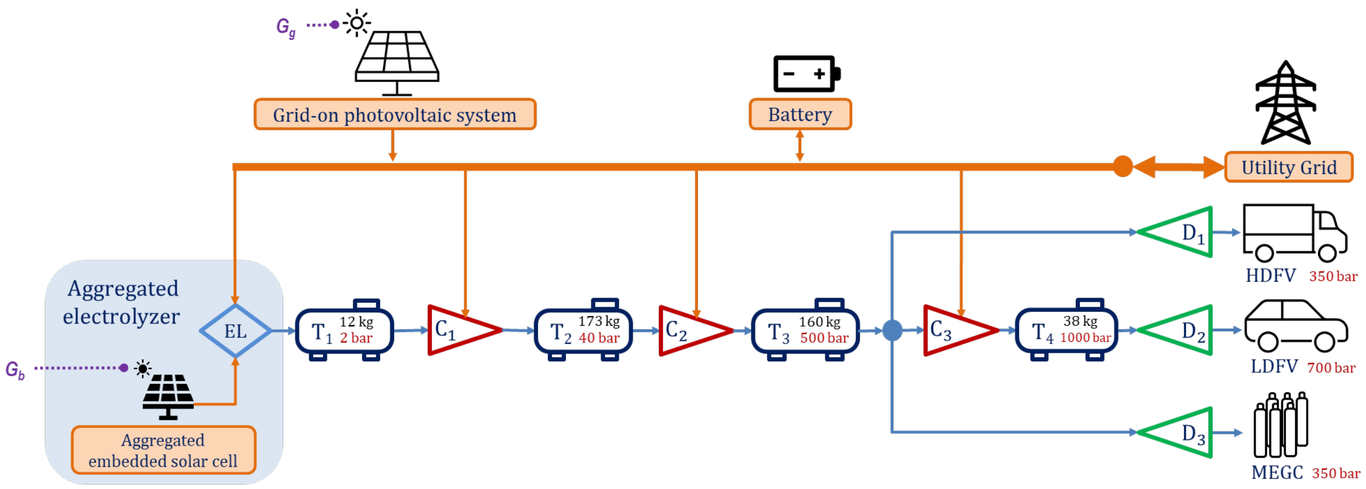
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Abstract

This work presents an **optimization-based controller**, specifically, a Chance-constrained Model Predictive Control (**CC-MPC**) strategy. Such a design is based on a **linear model** of a Hydrogen Refuelling Station (**HRS**) in which the **initial** Fuel Cell Electric Vehicles (**FCEVs**) on-board tank **pressure** at arrival is **uncertain**. The CC-MPC determines the electrolyzers and compressors operation as well as electricity consumption or sell to the utility grid.

Case of study



The **HRS studied** is based on a **real plant** under development at **Zaragoza**, **Spain**. The scheme at the left represents the main characteristics, which are the capability for **on-site hydrogen** electrolytic **production** by consuming energy from either **renewable sources** or the **utility grid**. Moreover, multiple compression and storage stages improve performance and operational flexibility. Multiple dispensers provide **cascaded refuelling** connecting a pressure vessel directly to the costumer **without** a **compressor**.

Linear discrete state-space model

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \mathbf{B}\boldsymbol{u}_k + \mathbf{B_d}\boldsymbol{d}_k$$

States

$$\boldsymbol{x}_k = \begin{pmatrix} x_{T1,k} \\ x_{T2,k} \\ x_{T3,k} \\ x_{T4,k} \\ x_{MEGC,k} \\ SOC_{BT,k} \end{pmatrix}$$

Inputs

$$\boldsymbol{u}_k = \begin{pmatrix} \gamma_{grid,k} \\ \gamma_{D3,k} \\ \gamma_{C1,k} \\ \gamma_{C2,k} \\ \gamma_{C3,k} \\ \gamma_{BT,k} \end{pmatrix}$$

Disturbances

$$\boldsymbol{d}_k = \begin{pmatrix} d_{Gb,k} \\ d_{HDFV,k} \\ d_{LDFV,k} \end{pmatrix}$$

State constraints $\mathcal{X} \triangleq \{\boldsymbol{x}_k \in \mathbb{R}^{n_x} \,|\, \mathbf{G}\boldsymbol{x}_k \leq \mathbf{g}\} \subset \mathbb{R}^{n_x}\ \forall k$

Input constraints $\mathcal{U} \triangleq \{\boldsymbol{u}_k \in \mathbb{R}^{n_u} \,|\, \mathbf{F}\boldsymbol{u}_k \leq \mathbf{f}\} \subset \mathbb{R}^{n_u}\ \forall k$

Stochastic hydrogen demand

Hydrogen demand

$$d_{(H/L)DFV,k} = \{0 \,|\, k \in \{m^{init}_{(H/L)DFV,k} = 0\}\} \cup \{m^{max}_{(H/L)DFV} - m^{init}_{(H/L)DFV,k} \,|\, k \in \{m^{init}_{(H/L)DFV,k} > 0\}\}$$

Max. mass; Initial mass

Initial mass

$$m^{init}_{(H/L)DFV,k} = \overline{m}^{init}_{(H/L)DFV,k} + \tilde{m}^{init}_{(H/L)DFV,k}$$

Mean value; Uncertain

Uncertainty assumed to follow distribution

$$m^{init}_{(H/L)DFV,k} \sim \mathcal{D}\left(\overline{m}^{init}_{(H/L)DFV,k}, \Sigma_{\tilde{m}^{init}_{(H/L)DFV,k}}\right)$$

Mean; Variance

Chance-constraint

To perform a **complete refuelling event**, the **pressure** vessel **connected** to the **dispenser must have** a **pressure** sufficiently **higher** than the **uncertain pressure** of the on-board **FCEV** tank upon **arrival**. This physical constraint $\mathcal{H}$ is formulated as a **chance constraint**. Hence, the probability of **constraint violation** is **bounded by** a **risk** factor:

$$\mathcal{H} \triangleq \left\{ \boldsymbol{x}_k \in \mathbb{R}^{n_x} \,|\, \begin{pmatrix} (x_{T3,k} - d_{HDFV,k})\mathrm{V}^{-1}_{T3} \\ (x_{T4,k} - d_{LDFV,k})\mathrm{V}^{-1}_{T4} \end{pmatrix} \geq \begin{pmatrix} (m^{init}_{HDFV,k} + d_{HDFV,k})\mathrm{V}^{-1}_{HDFV} - \varrho_{HDFV,k} \\ (m^{init}_{LDFV,k} + d_{LDFV,k})\mathrm{V}^{-1}_{LDFV} - \varrho_{LDFV,k} \end{pmatrix} \right\} \subset \mathbb{R}^{n_x}\ \forall k$$

Source tanks; FCEVs on-board tank; Slack

Probability

$$\mathbb{P}\left[\boldsymbol{x}_k \in \mathcal{H}\right] \geq 1 - \delta_d$$

Risk

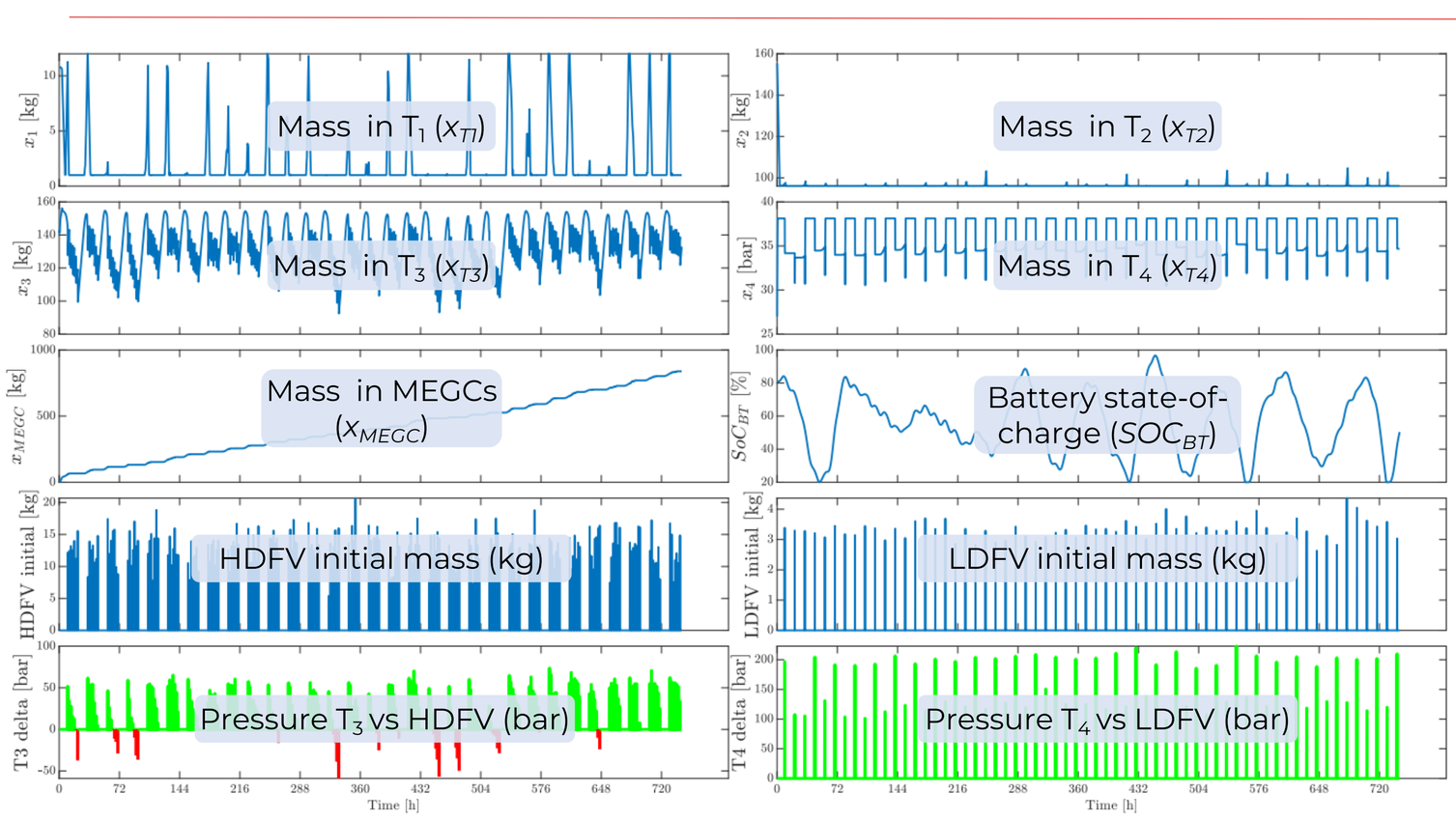
Slack $\quad [\varrho_{(H/L)DFV,k} \geq 0]\ \forall k \in \mathbb{N}^{H_p}_{+}$

Deterministic equivalent constraints **are derived**, being able to be **managed by** a **solver** in fixed-horizon optimization problems with receding horizon, such as in **Model Predictive Control**:

$$\mathcal{H}_{det} \triangleq \left\{ \boldsymbol{x}_k \in \mathbb{R}^{n_x} \,|\, \begin{pmatrix} x_{T3,k} - m^{max}_{HDFV}(1 + {}^{\mathrm{V}_{T3}}/_{\mathrm{V}_{HDFV}}) \\ x_{T4,k} - m^{max}_{LDFV}(1 + {}^{\mathrm{V}_{T4}}/_{\mathrm{V}_{LDFV}}) \end{pmatrix} \geq \begin{pmatrix} -\overline{m}^{init}_{HDFV,k} - \Phi^{-1}_{\mathcal{D}}(1 - {}^{\delta_d}/_{2H_p})\Sigma^{1/2}_{\tilde{m}^{init}_{HDFV,k}} - \varrho_{HDFV,k} \\ -\overline{m}^{init}_{LDFV,k} - \Phi^{-1}_{\mathcal{D}}(1 - {}^{\delta_d}/_{2H_p})\Sigma^{1/2}_{\tilde{m}^{init}_{LDFV,k}} - \varrho_{LDFV,k} \end{pmatrix} \right\} \subset \mathbb{R}^{n_x}\ \forall k \in \mathbb{N}^{H_p}_{+}$$

Inverse cumulative function; Slack

Chance-constrained Model Predictive Control results



Setup

- Prediction horizon (H_p): 10 hours
- Sampling time: 5 min
- Online forecasting
- Solver: GUROBI

Multi-objective function

- Maximize profits from electricity consumption/sell to the utility grid
- Minimize control inputs slew-rate
- Minimize slack variables (chance-constraint violation)

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